Optimising the efficiency of a drink can

Year 10 PSMT

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1 Introduction

1.1 Background Information

The relationship between volume and surface area is a critical consideration in many industries, with companies looking to optimise processes subject to constraints such as volume requirements. In the context of food and beverage, surface area is a key aspect of determining manufacturing costs for packaging. Beverage companies seek to minimise external surface area, thus minimising the packaging material required and cost. With such large volumes of product being sold every day, even minor adjustments can translate to significant savings and competitive advantages in the market.

1.2 Intent

This investigation will aim to find the most efficient cylindrical can enclosing a given volume. Through extensive mathematical analysis, this report will determine the optimal radius to minimise production cost for a canned beverage.

To approach this optimisation problem, the initial method of analysing empirical data with regression techniques will establish suitable models and approximations. A subsequent algebraic approach involving Lagrange multipliers will follow, providing exact values and verification of the answers obtained from the regressions.

2 Mathematical Analysis

2.1 Overview of Method

The can of choice will be a 500mL Monster Can. The can's volume (V) will be approximated as a cylinder and measured using radius (r) and height (h) of the extremities. Using the well known formula,

$$V(r, h) = \pi r^2 h$$

The dimensions of a standard 500mL Monster Can can be found below.

- $r = 3.3\text{cm}$
- $h = 15.7\text{cm}$



Figure 1: Dimensions of a 500mL Monster Can

The theoretical model provides a value of

$$\begin{aligned} V(r,h) &= \pi r^2 h \\ &= \pi(3.3\text{cm})^2(15.7\text{cm}) \\ &= 537.13\text{cm}^3 \\ &= 537.13\text{mL} \end{aligned}$$

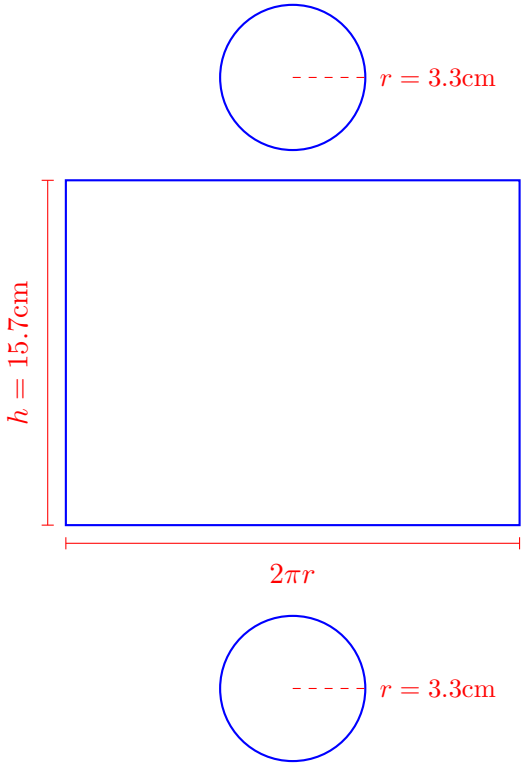


Figure 2: Net of a cylinder with height 15.7cm and radius 3.3cm

Observing Figure 2 above and using the well known formula for surface area of a cylinder,

$$S_1 = 2\pi rh$$
$$S_2 = 2\pi r^2$$

where S_1 is the curved wall surface area and S_2 is the sum of the areas of the top and bottom surfaces.

From this, it is possible to derive the cost $C(r, h)$ as a function of radius and height.

$$C(r, h) = 0.02S_1 + 0.06S_2$$
$$C(r, h) = (0.02)(2\pi rh) + (0.06)(2\pi r^2)$$

2.2 Observations

Observation 1. Almost all cans in supermarkets have radii falling within the range of 1-8cm.

The data will thus be restricted to points lying within this range to ensure that the values analysed are feasible and applicable to real world situations.

Observation 2. The can walls and top and bottom have different costs per unit cm^2.

This likely accounts for the different manufacturing costs due to different material thicknesses and construction of the lid.

Observation 3. The can is mostly cylindrical in shape, with grooves and bumps.

These serve functional purposes, such as stacking and forming a lid for the can. However, they dramatically complicate calculations due to the irregular shapes.

2.3 Assumptions

Assumption 1. The can may be approximated to be a perfect cylinder to a high degree of accuracy.

This assumption dramatically simplifies calculations for volume and surface area, whilst having a minimal impact on the accuracy of measurements as the can is approximately a cylinder.

Assumption 2. The can walls have negligible thickness.

Aluminium can walls generally have thicknesses of approximately 0.11mm, which may be considered negligible relative to its radius, height, and relevant dimensions (Incredible Can Strength, n.d.). This assumption also simplifies calculations as the can's measured volume can be taken to be equal to its capacity. The impact on the accuracy is also minimal as the can walls are generally a few tenths of a millimetre thick.

Assumption 3. The optimal radius exists within the range of 1-8cm.

It is assumed that beverage companies already create cans that are close to the most efficient radius, which all fall within the range of 1-8cm.

2.4 Identifying Relationships

The relationship between radius r, height h, volume $V(r,h)$ and cost $C(r,h)$ can be investigated with the use of software such as Excel by varying r whilst holding $V(r,h)$ constant.

Radius (cm)	Height (cm)	Volume (mL)	Top and Bottom (T/B) Surface Area (cm^2)	Curved Wall (W) Surface Area (cm^2)	Cost T/B (cents)	Cost W (cents)	Total Cost (cents)
3.3	15.7	537.1275208	68.423888	325.5318308	4.10543328	6.510636615	10.6160699
1	170.973	537.1275208	6.283185307	1074.255042	0.376991118	21.48510083	21.86209195

Figure 3: Extract of Excel Table of Values

Radius (cm)	Height (cm)	Volume (mL)	Top and Bottom (T/B) Surface Area (cm^2)	Curved Wall (W) Surface Area (cm^2)	Cost T/B (cents)	Cost W (cents)	Total Cost (cents)
3.3	15.7	=A2^2*B2*PI()	=2*PI()*A2^2	=2*PI()*A2*B2	=D2*0.06	=0.02*E2	=F2+G2
1	=C3/PI()/A3^2	537.127520762208	=2*PI()*A3^2	=2*PI()*A3*B3	=D3*0.06	=0.02*E3	=F3+G3

Figure 4: Extract of Excel Formulas

Note that h is given by simply rearranging $V(r,h)$ to give $h = \frac{V}{\pi r^2}$. The remaining formulas can be found in subsection 2.1. By applying the same formulas to several values for $r \in [1,8]$, the relationship can be examined through various regressions and their respective residuals. The full table with all values listed can be found in Appendix 6.1. As evident in the table, $C(r,h)$ approaches its minimum in the range $r \in [2.6, 3.4]$. The increment has thus been reduced to 0.1cm for this region of interest.

2.5 Regression Analysis

To analyse the data, linear, quadratic, and quartic regressions will be constructed with the use of software including Excel and Desmos. To evaluate how closely each regression fits, the coefficient of determination, R^2, and the quadratic mean, Q, of the residuals, e, will be taken. An R^2 value closer to 1 indicates better predictive accuracy.

To take the quadratic mean of the residuals, the formula

$$Q = \sqrt{\frac{\sum_{i=1}^{n} e_i^2}{n}}$$

will be used for n residuals. Since residuals are given by $e = A - T$ where A is the actual cost and T is the theoretical cost given by the regression, a lower Q value indicates better adherence to the data.

2.5.1 Linear Regression

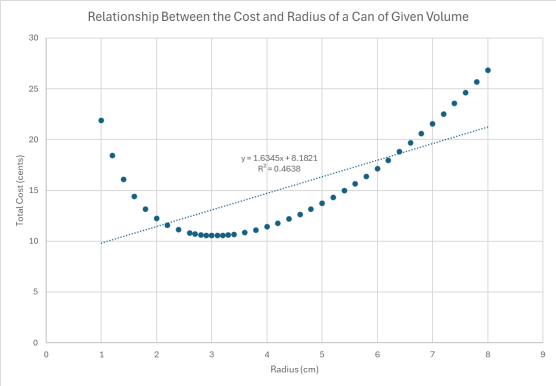


Figure 5: Linear Regression of Cost v. Radius

As shown in Figure 5, the linear regression is given by $C(r) = 1.6345r + 8.1821$. The low $R^2 = 0.4638$ value indicates high variation between the data and model.

The value $Q = 3.5554$ also indicates a high amount of deviation from the data. The details of this calculation can be found from Excel data in Appendix 6.2 and substitution into the formula for Q. Overall, the linear model is unfit to represent the data.

2.5.2 Quadratic Regression

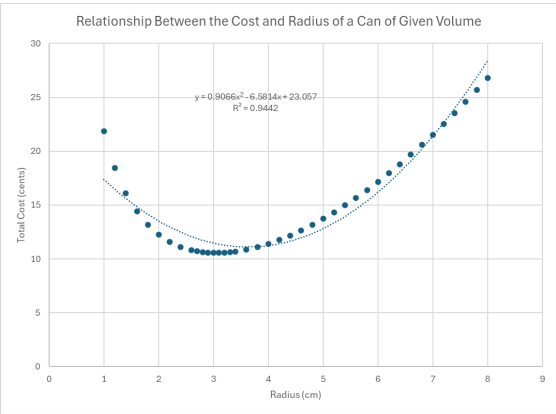


Figure 6: Quadratic Regression of Cost v. Radius

As shown in Figure 6, the quadratic regression is given by $C(r) = 0.9066r^2 - 6.5814r + 23.057$. The $R^2 = 0.9442$ value shows significant improvement over the linear model.

The value $Q = 1.1471$ also indicates a much lower amount of deviation from the data. The details of this calculation can be found from Excel data in Appendix 6.3 and

substitution into the formula for Q. Overall, the quadratic regression is a significant improvement to the model but still has room for improvement.

2.5.3 Quartic Regression

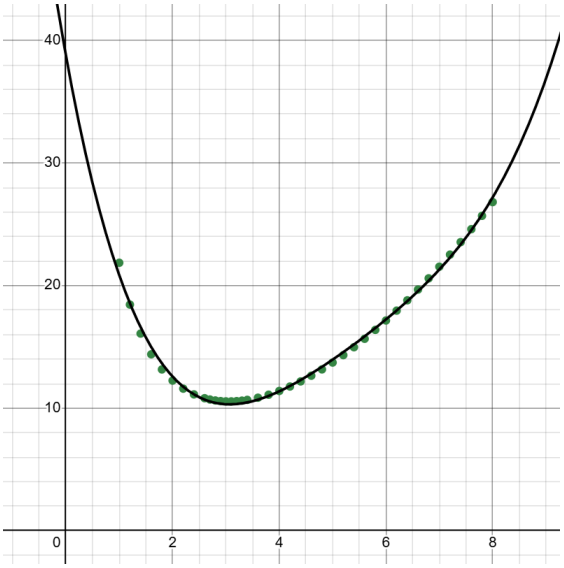


Figure 7: Quartic Regression of Cost v. Radius

As shown in Figure 7, the quartic regression is given by $C(r) = 0.0389915r^4 - 0.841617r^3 + 7.08341r^2 - 24.21664r + 38.78655$. The $R^2 = 0.9967$ value suggests that the quartic regression is an accurate approximation of $C(r)$ for $r \in [1, 8]$.

The value $Q = 0.2806$ displays minimal deviation from the data. The details of this calculation can be found from Desmos data in Appendix 6.4 and substitution into the formula for Q. Overall, the quartic regression is the best estimate for cost out of the three models.

Using the quartic model, the cost obtains its minimum at $r = 3.0727$ and $C(3.0727) = 10.3137$. The details of this calculation may be found in Appendix 6.4.3.

2.6 Exact Theoretical Answer and Verification

It can be shown that a radius of 3.054cm and a height of 18.33cm minimises the cost.

Proof. Begin by recalling the following functions and variables.

- $V(r, h) = \pi r^2 h$ determines the volume of the can (mL) as a function of its radius (cm) and height (cm).
- $C(r, h) = (0.02)(2\pi rh) + (0.06)(2\pi r^2)$ determines the cost of the can in cents as a function of its radius (cm) and height (cm).

The aim is to minimise the cost $C(r,h)$ subject to the constraint $V(r,h) = (3.3)^2(15.7)\pi$. By method of Lagrange multipliers,

$$\nabla V(r,h) = \lambda \nabla C(r,h)$$

where λ is the Lagrange multiplier.

Thus,

$$\begin{aligned}
\frac{\partial V}{\partial r} &= \lambda \frac{\partial C}{\partial r} \\
\frac{\partial V}{\partial h} &= \lambda \frac{\partial C}{\partial h} \\
V(r,h) &= (3.3)^2(15.7)\pi
\end{aligned}$$

and solving the above system of equations for r yields the minima.

Evaluating the derivatives,

$$\begin{aligned}
2\pi rh &= \lambda(0.02(2\pi h) + 0.06(4\pi r)) \\
rh &= 0.02\lambda h + 0.12\lambda r \\
h &= \frac{0.12\lambda r}{r - 0.02\lambda} \qquad (1) \\
\pi r^2 &= \lambda(0.02(2\pi r)) \\
r &= 0.04\lambda \qquad (2)
\end{aligned}$$

By substituting (2) into (1),

$$\begin{aligned}
h &= \frac{0.12\lambda(0.04\lambda)}{0.04\lambda - 0.02\lambda} \\
h &= 0.24\lambda \qquad (3)
\end{aligned}$$

Now substituting (2) and (3) into $V(r,h) = (3.3)^2(15.7)\pi$,

$$\begin{aligned}
\pi(0.04\lambda)^2(0.24\lambda) &= (3.3)^2(15.7)\pi \\
\lambda &= \sqrt[3]{\frac{(3.3)^2(15.7)}{(0.04)^2(0.24)}}
\end{aligned}$$

and from (2) and (3),

$$\begin{aligned}
r &= 0.04\sqrt[3]{\frac{(3.3)^2(15.7)}{(0.04)^2(0.24)}} \\
r &= 3.054\text{cm} \\
h &= 0.24\sqrt[3]{\frac{(3.3)^2(15.7)}{(0.04)^2(0.24)}} \\
h &= 18.33\text{cm}
\end{aligned}$$

Although the method of Lagrange multipliers provides either the maximum or minimum of a function subject to a constraint, it is important to verify that this is the minimum. This is easy to check by comparing with any of the values found in Appendix 6.1.

Substituting these values into $C(r, h)$ yields

$$\begin{aligned} C(r, h) &= 2\pi(0.02)(3.054)(18.33) + 2\pi(0.06)(3.054)^2 \\ &= 10.55¢ \end{aligned}$$

QED

Alternatively, $C(r, h)$ can be modelled in Desmos as follows

$$V(r, h) = \pi r^2 h$$
$$h = \frac{V}{\pi r^2}$$

Substituting into $C(r, h)$,

$$C(r) = (0.02)(2\pi r)\left(\frac{V}{\pi r^2}\right) + (0.06)(2\pi r^2)$$

The only variable is r since V is constant so the cost function is a function of radius only. Thus, it is possible to graph cost against radius.

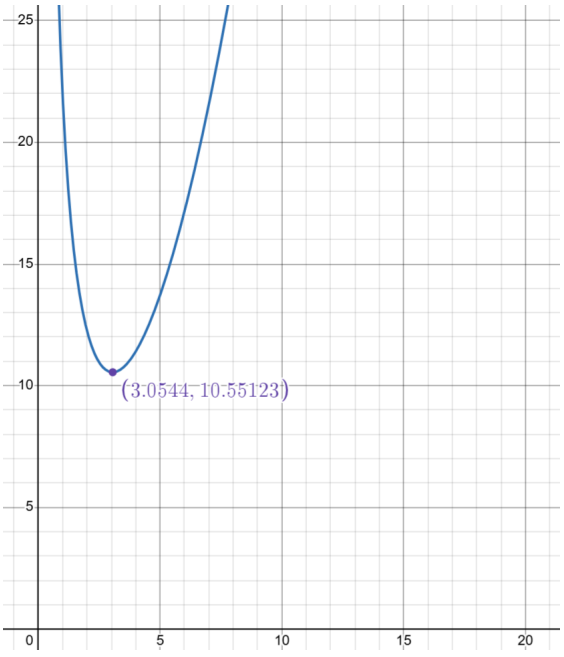


Figure 8: Cost v. Radius

The minimum according to the Desmos model in Figure 8 is clearly in agreement with the value produced by the method of Lagrange multipliers.

3 Evaluation

Ultimately, the quartic model predicted a minimum cost of 10.31¢ with a radius of 3.073cm. An algebraic approach revealed that the optimal radius was 3.054cm with a cost of 10.55¢. Thus, although the quartic regression represents the data fairly accurately, there are still deviations from the theoretical values. The final dimensions obtained were similar to those of the original 500mL Monster can, with the radius only differing by 0.246cm. With the original can costing 10.62¢, the modifications suggested by this investigation would result in a 1.04% cost reduction. Although minor, this difference results in significant savings in large scale production with a company such as Monster.

3.1 Assessment of Initial Assumptions

Many of the simplifying assumptions made in the calculations also affected the validity of the final result. The approximation of the can as a perfect cylinder was unrealistic, neglecting the grooves in the can. The assumption that the can walls had negligible thickness was reasonable, but also unrealistic. The can wall thickness would reduce the capacity. These two factors combined likely contributed to the difference between the calculated volume and listed 500mL capacity.

Although the range of $r \in [1, 8]$ could have been expanded, this had little effect on the investigation, with almost all cans falling within this range. The algebraic approach also avoids this constraint, showing that the cost values in this range exceed the minimum obtained.

3.2 Further Limitations

Although packaging cost was the primary factor considered throughout this investigation, many other considerations are necessary for beverage companies. For instance, the curved walls are often utilised for marketing purposes and sufficient surface area is required. Another restriction would be storage considerations, with companies seeking to maximise product stored within a given volume.

The given condition that the top and bottom surfaces are $0.06¢/\text{cm}^2$ and curved walls are $0.02¢/\text{cm}^2$ does not necessarily hold true in all real world applications. The real production costs of these would have to be determined by individual companies to ensure an accurate model. These factors were beyond the scope of this investigation.

3.3 Extensions

For future work on this topic, a more exact approach can be adopted, using the exact dimensions and shape of the can rather than relying on approximations. Furthermore, the can wall thickness can be accounted for and thus a more accurate value for volume. The packing efficiency should also be considered as transportation and storage are costs associated with the production of beverages.

4 Conclusion

In conclusion, this investigation optimised the radius of a 500mL Monster can, by minimising the cost. After analysis of various regressions with Excel and Desmos and an algebraic approach involving Lagrange multipliers, it was found that the optimal radius was 3.054cm with a cost of 10.55¢.

Overall, this investigation could be improved by incorporating the aforementioned considerations. In real world applications, more factors contribute to the cost and issues surrounding usability and sustainability exist. Nonetheless, with the appropriate techniques, they may also be optimised with the use of mathematical analysis.

5 References

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3. Monster Energy Sports & Energy Drinks Fizzy Drinks, drink, food, sugar, monster Energy png — PNGWing. (2025). Retrieved August 13, 2025, from Pngwing.com website: https://www.pngwing.com/en/free-png-setfa

6 Appendices

6.1 Full Data Table

Radius (cm)	Height (cm)	Volume (mL)	Top and Bottom (T/B) Surface Area (cm^2)	Curved Wall (W) Surface Area (cm^2)	Cost T/B (cents)	Cost W (cents)	Total Cost (cents)
1	170.973	537.1275208	6.283185307	1074.255042	0.376991118	21.48510083	21.86209195
1.2	118.73125	537.1275208	9.047786842	895.2125346	0.542867211	17.90425069	18.4471179
1.4	87.231122	537.1275208	12.3150432	767.3250297	0.738902592	15.34650059	16.08540319
1.6	66.786328	537.1275208	16.08495439	671.409401	0.965097263	13.42818802	14.39328528
1.8	52.769444	537.1275208	20.3575204	596.8083564	1.221451224	11.93616713	13.15761835
2	42.74325	537.1275208	25.13274123	537.1275208	1.507964474	10.74255042	12.25051489
2.2	35.325	537.1275208	30.41061689	488.2977461	1.824637013	9.765954923	11.59059194
2.4	29.682813	537.1275208	36.19114737	447.6062673	2.171468842	8.952125346	11.12359419
2.6	25.291864	537.1275208	42.47433268	413.175016	2.548459961	8.263500319	10.81196028
2.7	23.453086	537.1275208	45.80442089	397.8722376	2.746265253	7.957444752	10.70371001
2.8	21.807781	537.1275208	49.26017281	383.6625148	2.955610368	7.673250297	10.62886067
2.9	20.329727	537.1275208	52.84158843	370.4327729	3.170495306	7.408655459	10.57915076
3	18.997	537.1275208	56.54866776	358.0850138	3.392920066	7.161700277	10.55462034
3.1	17.791155	537.1275208	60.3814108	346.5338844	3.622884648	6.930677687	10.55356234
3.2	16.696582	537.1275208	64.33981755	335.7047005	3.860389053	6.71409401	10.57448306
3.3	15.7	537.1275208	68.423888	325.5318308	4.10543328	6.510636615	10.6160699
3.4	14.790052	537.1275208	72.63362215	315.9573652	4.358017329	6.319147303	10.67716463
3.6	13.192361	537.1275208	81.43008158	298.4041782	4.885804895	5.968083564	10.85388846
3.8	11.840235	537.1275208	90.72919584	282.6986951	5.44375175	5.653973903	11.09772565
4	10.685813	537.1275208	100.5309649	268.5637604	6.031857895	5.371275208	11.4031331
4.2	9.6923469	537.1275208	110.8353888	255.7750099	6.650123329	5.115500198	11.76562353
4.4	8.83125	537.1275208	121.6424675	244.1488731	7.298548053	4.882977461	12.18152551
4.6	8.0800095	537.1275208	132.9522011	233.5337047	7.977132066	4.670674094	12.64780616
4.8	7.4207031	537.1275208	144.7645895	223.8031337	8.685875369	4.476062673	13.16193804
5	6.83892	537.1275208	157.0796327	214.8510083	9.424777961	4.297020166	13.72179813
5.2	6.322966	537.1275208	169.8973307	206.587508	10.19383984	4.13175016	14.32559
5.4	5.8632716	537.1275208	183.2176836	198.9361188	10.99306101	3.978722376	14.97178339
5.6	5.4519452	537.1275208	197.0406912	191.8312574	11.82244147	3.836625148	15.65906662
5.8	5.0824316	537.1275208	211.3663537	185.2163865	12.68198122	3.704327729	16.38630895
6	4.74925	537.1275208	226.1946711	179.0425069	13.57168026	3.580850138	17.1525304
6.2	4.4477888	537.1275208	241.5256432	173.2669422	14.49153859	3.465338844	17.95687744
6.4	4.1741455	537.1275208	257.3592702	167.8523502	15.44155621	3.357047005	18.79860322
6.6	3.925	537.1275208	273.695552	162.7659154	16.42173312	3.255318308	19.67705143
6.8	3.697513	537.1275208	290.5344886	157.9786826	17.43206932	3.159573652	20.59164297
7	3.4892449	537.1275208	307.8760801	153.4650059	18.4725648	3.069300119	21.54186492
7.2	3.2980903	537.1275208	325.7203263	149.2020891	19.54321958	2.984041782	22.52726136
7.4	3.1222243	537.1275208	344.0672274	145.1696002	20.64403365	2.903392004	23.54742565
7.6	2.9600589	537.1275208	362.9167833	141.3493476	21.775007	2.826986951	24.60199395
7.8	2.8102071	537.1275208	382.2689941	137.7250053	22.93613965	2.754500106	25.69063975
8	2.6714531	537.1275208	402.1238597	134.2818802	24.12743158	2.685637604	26.81306918

6.2 Linear Regression Analysis

6.2.1 Residuals

Regression Prediction	e	e^2
9.8166	12.045492	145.0939
10.1435	8.3036179	68.95007
10.4704	5.6150032	31.52826
10.7973	3.5959853	12.93111
11.1242	2.0334184	4.13479
11.4511	0.7994149	0.639064
11.778	-0.187408	0.035122
12.1049	-0.981306	0.962961
12.4318	-1.61984	2.623881
12.59525	-1.88954	3.570361
12.7587	-2.129839	4.536216
12.92215	-2.342999	5.489645
13.0856	-2.53098	6.405858
13.24905	-2.695488	7.265654
13.4125	-2.838017	8.05434
13.57595	-2.95988	8.76089
13.7394	-3.062235	9.377285
14.0663	-3.212412	10.31959
14.3932	-3.295474	10.86015
14.7201	-3.316967	11.00227
15.047	-3.281376	10.76743
15.3739	-3.192374	10.19125
15.7008	-3.052994	9.320771
16.0277	-2.865762	8.212592
16.3546	-2.632802	6.931646
16.6815	-2.35591	5.550312
17.0084	-2.036617	4.147807
17.3353	-1.676233	2.809758
17.6622	-1.275891	1.627898
17.9891	-0.83657	0.699849
18.316	-0.359123	0.128969
18.6429	0.1557032	0.024243
18.9698	0.7072514	0.500205
19.2967	1.294943	1.676877
19.6236	1.9182649	3.67974
19.9505	2.5767614	6.639699
20.2774	3.2700256	10.69307
20.6043	3.997694	15.98156
20.9312	4.7594398	22.65227
21.2581	5.5549692	30.85768

6.2.2 Residual Formulas

Regression Prediction	e	e^2
=1.6345*(A2)+8.1821	=H2-I2	=J2^2
=1.6345*(A3)+8.1821	=H3-I3	=J3^2
=1.6345*(A4)+8.1821	=H4-I4	=J4^2
=1.6345*(A5)+8.1821	=H5-I5	=I5^2
=1.6345*(A6)+8.1821	=H6-I6	=J6^2
=1.6345*(A7)+8.1821	=H7-I7	=J7^2
=1.6345*(A8)+8.1821	=H8-I8	=J8^2
=1.6345*(A9)+8.1821	=H9-I9	=J9^2
=1.6345*(A10)+8.1821	=H10-I10	=J10^2
=1.6345*(A11)+8.1821	=H11-I11	=J11^2
=1.6345*(A12)+8.1821	=H12-I12	=J12^2
=1.6345*(A13)+8.1821	=H13-I13	=J13^2
=1.6345*(A14)+8.1821	=H14-I14	=J14^2
=1.6345*(A15)+8.1821	=H15-I15	=I15^2
=1.6345*(A16)+8.1821	=H16-I16	=I16^2
=1.6345*(A17)+8.1821	=H17-I17	=J17^2
=1.6345*(A18)+8.1821	=H18-I18	=J18^2
=1.6345*(A19)+8.1821	=H19-I19	=J19^2
=1.6345*(A20)+8.1821	=H20-I20	=J20^2
=1.6345*(A21)+8.1821	=H21-I21	=J21^2
=1.6345*(A22)+8.1821	=H22-I22	=J22^2
=1.6345*(A23)+8.1821	=H23-I23	=J23^2
=1.6345*(A24)+8.1821	=H24-I24	=J24^2
=1.6345*(A25)+8.1821	=H25-I25	=J25^2
=1.6345*(A26)+8.1821	=H26-I26	=J26^2
=1.6345*(A27)+8.1821	=H27-I27	=J27^2
=1.6345*(A28)+8.1821	=H28-I28	=J28^2
=1.6345*(A29)+8.1821	=H29-I29	=J29^2
=1.6345*(A30)+8.1821	=H30-I30	=J30^2
=1.6345*(A31)+8.1821	=H31-I31	=J31^2
=1.6345*(A32)+8.1821	=H32-I32	=J32^2
=1.6345*(A33)+8.1821	=H33-I33	=J33^2
=1.6345*(A34)+8.1821	=H34-I34	=J34^2
=1.6345*(A35)+8.1821	=H35-I35	=J35^2
=1.6345*(A36)+8.1821	=H36-I36	=J36^2
=1.6345*(A37)+8.1821	=H37-I37	=J37^2
=1.6345*(A38)+8.1821	=H38-I38	=J38^2
=1.6345*(A39)+8.1821	=H39-I39	=J39^2
=1.6345*(A40)+8.1821	=H40-I40	=J40^2
=1.6345*(A41)+8.1821	=H41-I41	=J41^2

6.3 Quadratic Regression Analysis

6.3.1 Residuals

Regression Prediction	e	e^2
17.3822	4.479892	20.0694
16.464824	1.982294	3.92949
15.619976	0.465427	0.21662
14.847656	-0.454371	0.20645
14.147864	-0.990246	0.98059
13.5206	-1.270085	1.61312
12.965864	-1.375272	1.89137
12.483656	-1.360062	1.84977
12.073976	-1.262016	1.59268
11.896334	-1.190624	1.41759
11.736824	-1.107963	1.22758
11.595446	-1.016295	1.03286
11.4722	-0.91758	0.84195
11.367086	-0.813524	0.66182
11.280104	-0.705621	0.4979
11.211254	-0.595184	0.35424
11.160536	-0.483371	0.23365
11.113496	-0.259608	0.0674
11.138984	-0.041258	0.0017
11.237	0.166133	0.0276
11.407544	0.35808	0.12822
11.650616	0.53091	0.28186
11.966216	0.68159	0.46457
12.354344	0.807594	0.65221
12.815	0.906798	0.82228
13.348184	0.977406	0.95532
13.953896	1.017887	1.03609
14.632136	1.026931	1.05459
15.382904	1.003405	1.00682
16.2062	0.94633	0.89554
17.102024	0.854853	0.73077
18.070376	0.728227	0.53031
19.111256	0.565795	0.32012
20.224664	0.366979	0.13467
21.4106	0.131265	0.01723
22.669064	-0.141803	0.02011
24.000056	-0.45263	0.20487
25.403576	-0.801582	0.64253
26.879624	-1.188984	1.41368
28.4282	-1.615131	2.60865

6.3.2 Residual Formulas

Regression Prediction	e	e^2
=0.9066*(A2)^2-6.5814*(A2)+23.057	=H2-I2	=J2^2
=0.9066*(A3)^2-6.5814*(A3)+23.057	=H3-I3	=J3^2
=0.9066*(A4)^2-6.5814*(A4)+23.057	=H4-I4	=J4^2
=0.9066*(A5)^2-6.5814*(A5)+23.057	=H5-I5	=J5^2
=0.9066*(A6)^2-6.5814*(A6)+23.057	=H6-I6	=J6^2
=0.9066*(A7)^2-6.5814*(A7)+23.057	=H7-I7	=J7^2
=0.9066*(A8)^2-6.5814*(A8)+23.057	=H8-I8	=J8^2
=0.9066*(A9)^2-6.5814*(A9)+23.057	=H9-I9	=J9^2
=0.9066*(A10)^2-6.5814*(A10)+23.057	=H10-I10	=J10^2
=0.9066*(A11)^2-6.5814*(A11)+23.057	=H11-I11	=J11^2
=0.9066*(A12)^2-6.5814*(A12)+23.057	=H12-I12	=J12^2
=0.9066*(A13)^2-6.5814*(A13)+23.057	=H13-I13	=J13^2
=0.9066*(A14)^2-6.5814*(A14)+23.057	=H14-I14	=J14^2
=0.9066*(A15)^2-6.5814*(A15)+23.057	=H15-I15	=J15^2
=0.9066*(A16)^2-6.5814*(A16)+23.057	=H16-I16	=J16^2
=0.9066*(A17)^2-6.5814*(A17)+23.057	=H17-I17	=J17^2
=0.9066*(A18)^2-6.5814*(A18)+23.057	=H18-I18	=J18^2
=0.9066*(A19)^2-6.5814*(A19)+23.057	=H19-I19	=J19^2
=0.9066*(A20)^2-6.5814*(A20)+23.057	=H20-I20	=J20^2
=0.9066*(A21)^2-6.5814*(A21)+23.057	=H21-I21	=J21^2
=0.9066*(A22)^2-6.5814*(A22)+23.057	=H22-I22	=J22^2
=0.9066*(A23)^2-6.5814*(A23)+23.057	=H23-I23	=J23^2
=0.9066*(A24)^2-6.5814*(A24)+23.057	=H24-I24	=J24^2
=0.9066*(A25)^2-6.5814*(A25)+23.057	=H25-I25	=J25^2
=0.9066*(A26)^2-6.5814*(A26)+23.057	=H26-I26	=J26^2
=0.9066*(A27)^2-6.5814*(A27)+23.057	=H27-I27	=J27^2
=0.9066*(A28)^2-6.5814*(A28)+23.057	=H28-I28	=J28^2
=0.9066*(A29)^2-6.5814*(A29)+23.057	=H29-I29	=J29^2
=0.9066*(A30)^2-6.5814*(A30)+23.057	=H30-I30	=J30^2
=0.9066*(A31)^2-6.5814*(A31)+23.057	=H31-I31	=J31^2
=0.9066*(A32)^2-6.5814*(A32)+23.057	=H32-I32	=J32^2
=0.9066*(A33)^2-6.5814*(A33)+23.057	=H33-I33	=J33^2
=0.9066*(A34)^2-6.5814*(A34)+23.057	=H34-I34	=J34^2
=0.9066*(A35)^2-6.5814*(A35)+23.057	=H35-I35	=J35^2
=0.9066*(A36)^2-6.5814*(A36)+23.057	=H36-I36	=J36^2
=0.9066*(A37)^2-6.5814*(A37)+23.057	=H37-I37	=J37^2
=0.9066*(A38)^2-6.5814*(A38)+23.057	=H38-I38	=J38^2
=0.9066*(A39)^2-6.5814*(A39)+23.057	=H39-I39	=J39^2
=0.9066*(A40)^2-6.5814*(A40)+23.057	=H40-I40	=J40^2
=0.9066*(A41)^2-6.5814*(A41)+23.057	=H41-I41	=J41^2

6.4 Quartic Regression Analysis

6.4.1 Residuals

x_1	y_1	e_1^2
1	21.86209195	1.0229194
1.2	18.4471179	0.01126094
1.4	16.08540319	0.272206
1.6	14.39328528	0.34627413
1.8	13.15761835	0.24034139
2	12.25051489	0.10714969
2.2	11.59059194	0.024004277
29 more rows Show all		
7.4	23.54742565	0.039004002
7.6	24.60199395	0.0079977772
7.8	25.69063975	0.0094627793
8	26.81306918	0.14452797

Quartic Regression

EQUATION

$$y = 0.0389915x^4 - 0.841617x^3 + 7.08341x^2 - 24.21664x + 38.78655$$

STATISTICS

$R^2 = 0.9967$

RESIDUALS

e_1

6.4.2 Residual Formulas

$$\sqrt{\frac{\text{total}\left(e_1^2\right)}{40}} = 0.280555139122$$

6.4.3 Quartic Minimum Calculation

The function $C = 0.03899r^4 - 0.8416r^3 + 7.0834r^2 - 24.2166r + 38.7866$, obtains its minima at $\frac{\mathrm{d}C}{\mathrm{d}r} = 0$

$$\frac{\mathrm{d}C}{\mathrm{d}r} = 0.1560r^3 - 2.5249r^2 + 14.1668r - 24.2166.$$

For the resulting cubic equation,

$$0.1560r^3 - 2.5249r^2 + 14.1668r - 24.2166 = 0.$$

Let $a = 0.1560$, $b = -2.5249$, $c = 14.1668$, $d = -24.2166$. To solve, depress the cubic by dividing by a.

$$r^3 + Ar^2 + Br + C = 0$$

where $A = b/a = -16.1885$, $B = c/a = 90.8327$, $C = d/a = -155.2687$. Substitute $r = z - A/3 = z + 5.3962$ to obtain the depressed cubic

$$z^3 + pz + q = 0,$$

where

$$p = \frac{3a \cdot c - b^2}{3a^2} = 3.4772$$
$$q = \frac{2b^3 - 9a \cdot b \cdot c + 27a^2 \cdot d}{27a^3} = 20.6229.$$

The discriminant is

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 = 107.8837.$$

Since $\Delta > 0$, there is one real root.

$$s_1 = -\frac{q}{2} + \sqrt{\Delta} = 0.0752$$
$$s_2 = -\frac{q}{2} - \sqrt{\Delta} = -20.6982.$$

The real cube roots are

$$u = \sqrt[3]{s_1} = 0.4221,$$
$$v = \sqrt[3]{s_2} = -2.7456.$$

Then,

$$z = u + v = -2.3235$$
$$r = z - \frac{A}{3} = 3.0727$$

To check that this is the minimum, compute the second derivative.

$$\frac{\mathrm{d}^2 C}{\mathrm{d}r^2} = 0.4679r^2 - 5.0497r + 14.1668.$$

At $r = 3.0727$,

$$\frac{\mathrm{d}^2 C}{\mathrm{d}r^2} = 3.0683 > 0,$$

Since $\frac{\mathrm{d}^2 C}{\mathrm{d}r^2} > 0$, the function is convex at this point and the minimum is thus obtained.

$$C(3.0727) = 10.3137.$$

End of Paper